

Preparation for Postsecondary Education of Transition Aged Youth with Intellectual
Disability and Autism Spectrum Disorder in the United States: An Analysis of Data
from the National Longitudinal Transition Study 2012

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Abstract

Youth with intellectual disability (defined as “a condition characterized by significant limitations in both intellectual functioning and adaptive behavior that originates before the age of 22, American Association on Intellectual and Developmental Disabilities, 2023) and autism spectrum disorder increasingly access postsecondary education in many countries around the world. To ensure students are ready to access these options, preparation for postsecondary education must be part of their transition services. This study examines the postsecondary education preparation experiences of youth with intellectual disability and autism using the NLTS 2012 dataset to identify the extent to which these youth are accessing preparation activities and if this preparation differs from youth in other groups. Findings indicate few differences between the postsecondary education preparation of youth with intellectual disability and autism and autism without intellectual disability but highlight substantial differences in several preparation activities when compared with youth in other groups.

Keywords: intellectual disability, autism spectrum disorder, postsecondary education, transition.

Across the globe, an increasing number of nations have been developing access to postsecondary education for students with intellectual disability and autism (O'Brien and Bonati, 2019; Plaute, 2022). In Europe, postsecondary education options for students with intellectual disability and autism are available in Austria, Germany, Iceland, Ireland, Italy, Switzerland, and Spain (Plaute, 2022). Other countries including Australia, Canada, and Portugal have also begun to develop or grow postsecondary education options for this population of students (O'Brien and Bonati, 2019). In many of these countries, efforts are focused on a specific university and are not widespread (Grigal et al., in press). However, in the United States, access to postsecondary education is more widespread as it is supported through government funding and legislation (i.e., Higher Education Opportunities Act [HEOA] of 2008, Workforce Innovation Opportunity Act of 2014).

The number of postsecondary education programs for students with intellectual disability, and often co-occurring autism, in the U.S. is now over 300 (Think College, 2023). Program availability and student enrollment has grown substantially in the past decade (Grigal et al., 2022). One catalyst for this growth has been a government funded program called the Transition and Postsecondary Education for Students with Intellectual Disabilities (TPSID) model demonstration program. The TPSID model demonstration program has provided competitive grant funds to institutions of higher education to support development or enhancement of programs for students with intellectual disability. TPSID funds have been received by over 100 postsecondary education institutions in 31 states since 2010 (Think College National Coordinating Center, 2021).

Preparing students, including students with intellectual disability and autism, to access postsecondary education is critical to their success. In the U.S., students with disabilities are

entitled to a free and appropriate public education until the age of 18, but in many states, students with intellectual disability and autism can continue to attend high school and receive special education services through the age of 21 or 22 years of age. These high school¹ experiences seek to prepare students for two potential critical outcomes: attending postsecondary education (referred to as college²) and pursuing a career.

For students with disabilities, activities that prepare them to transition into postsecondary education or into a career should be a part of the transition plan in the Individualized Education Program (IEP) as mandated by the Individuals with Disabilities Education Act (IDEA, 2004). However, not all students receiving special education services in the country have the same access to transition planning and postsecondary education preparation. Though 69% of youth across all IDEA disability categories report some level of involvement in developing the transition plan for their annual IEP meetings, students with intellectual disability and students with autism spectrum disorder (referred to as autism hereafter) have also been found to be less likely to provide input in their IEP and transition planning than youth of other disability groups (Lipscomb et al., 2017).

Likewise, 54% of parents of autistic youth report receiving less information from the school related to postsecondary education, employment, and independent living options than do parents of youth in most other disability groups (except for youth with deaf-blindness and traumatic brain injury who have the lowest reported rates; Lipscomb et al., 2017). Additionally, earlier studies indicated youth with intellectual disability and autism do not engage in academic

¹ Secondary education, referred to as high school in the US, typically is provided to students from age 14-18, but some students with intellectual disability and autism continue their secondary education through their 21st year.

² College is the generic term for any post-secondary undergraduate education in the U.S. Americans attend “college” after high school, typically after age 18. Colleges and universities both offer undergraduate degrees.

or extracurricular activities to prepare for postsecondary education at the same rate as peers with other disabilities (Grigal et al., 2011; Weiss and Burnham Riosa, 2015; Zeedyk, et al., 2016).

Postsecondary education access is especially critical for students with intellectual disability and autism as it can have a significant impact on both employment outcomes and wages. Multiple studies conducted using the Rehabilitation Services Administration's Case Service Report (RSA-911) dataset concluded vocational rehabilitation (VR) clients with intellectual disability and autism who access postsecondary education are more likely to be employed and earn higher wages than those who do not access postsecondary education (Miller et al., 2019; Sannicandro et al., 2018; Smith et al., 2018). Similarly, Shattuck et al. (2012) found that 55% of autistic students who participated in postsecondary education were employed after high school. Students with intellectual disability and autism enrolled in colleges and universities receiving TPSID grant funds have also demonstrated positive employment outcomes. Students' employment rates up to three years after graduating have been 60% or higher (Grigal et al., 2021b), which is significantly higher than the fifteen percent employment rate reported in 2021 for adults with intellectual and developmental disabilities in the general U.S. population (National Core Indicators, 2022).

Postsecondary education preparation during high school is especially vital for students with intellectual disability and autism who face additional barriers to accessing postsecondary education. A variety of school-based practices can help students with disabilities better prepare for postsecondary education (Mazzotti et al., 2016; Mazzotti et al., 2021; Test et al., 2009). Some of these practices relate to course work such as taking career preparation classes or classes that offer postsecondary education credit (also referred to as dual enrollment courses). Other activities involve attending postsecondary education fairs or visiting postsecondary education

institutions. Finally, there are necessary activities required by all students who seek postsecondary education in the U.S. including taking entrance or placement tests, filling out financial aid forms, and completing applications.

Given the broadening opportunities for students with intellectual disability and autism to access postsecondary education, it is important to examine the postsecondary education preparation activities offered during their secondary school experiences. Previous studies reflected students with intellectual disability and autism in the past had poorer preparation than their peers with and without disabilities. The current study used more recent data from the National Longitudinal Transition Study 2012 (NLTS 2012) to identify the extent to which youth with intellectual disability and autism are accessing postsecondary education preparation activities. It also explores if this preparation differs by student characteristics including autistic youth without intellectual disability, youth with other disabilities, youth with 504 plans but not IEPs, and youth with neither 504 plans nor IEPs. It is important to note that although 504 plans are like IEPs in that they provide students with disabilities access to education without discrimination (see: Section 504 of the Rehabilitation Act of 1973), students with 504 plans are regarded as a separate group because the types of services they are provided differ. Students with 504 plans are primarily supported through accommodations that enable access (e.g., changes to environment or instruction) whereas students with IEPs receive continuous special education services in academic and non-academic settings (Pacer Center, 2023).

The findings of this study can help the field of transition and secondary special education better understand who is and is not getting access to postsecondary education preparation activities and help identify recommendations that might improve future postsecondary education preparation for these youth. The study addressed the following two research questions:

1. To what extent are youth with intellectual disability and autism accessing postsecondary education preparation?
2. Is the access to postsecondary education preparation for youth with intellectual disability and autism different from youth in other groups (i.e., youth with autism but not intellectual disability, other disabilities, 504 plans, and neither 504 plans nor IEPs)?

Methods

The NLTS 2012 dataset is the third and most recent data set in the National Longitudinal Transition Study series studying the characteristics and experiences of youth with disabilities in the U.S. This updated version of the dataset is the first to allow for direct comparisons of youth with and without disabilities who were thirteen to twenty-one years old and in grades seven through twelve during the 2011-12 school year. The NLTS 2012 dataset is the most recent national representation of transition preparation practices experienced by most youth with disabilities in the US. Using the NLTS 2012 dataset, recent studies have enhanced our understanding of other critical components of transition planning such as parent involvement (Qian et al., 2020; Zeng et al., 2022) and student participation in transition planning (Bouck et al., 2021; Davenport et al., 2022).

Using the complete NLTS 2012 dataset, we conducted a secondary data analysis of a nationally representative sample of ~12,000 youth with and without disabilities. Of these students, 9,600 were students with an IEP, 600 students had 504 plans but not IEPs, and 1,800 students had neither IEPs nor 504 plans (note: demographic figures were rounded to the closest decimal in compliance with the Institute of Education Sciences). The students with an IEP included youth from each of the thirteen disability categories listed under IDEA (2004): autism, deaf-blindness, deafness, emotional disturbance, hearing impairment, intellectual disability,

multiple disabilities, orthopedic impairment, other health impairment, specific learning disability, speech or language impairment, traumatic brain injury, and visual impairment. The NLTS 2012 study was carried out by Mathematica Policy Research. Data were collected through computer-assisted interviews with the students and their parents as well as administrative records from the participating school districts. More details about the research methodology are available in Bloomenthal et al. (2017) and Burghardt et al. (2017).

Youth characteristics

This paper focuses on five groups of youth in high school who were sixteen years or older: a) Youth with intellectual disability who had an IEP including autistic youth who had also been identified as having intellectual disability (intellectual disability and autism, $n = 1,240$); b) autistic youth with IEPs who did not have a dual diagnosis of intellectual disability ($n = 680$); c) youth with other disabilities with IEPs ($n = 6,230$); d) youth with 504 plans but not IEPs ($n = 490$; note: the disability of youth with 504 plans is not specified in the dataset so this figure could also include youth with intellectual disability or autism); and e) youth with neither 504 plans nor IEPs ($n = 1,360$). We focused on high school youth who were between sixteen and twenty-one years old to include only youth for whom postsecondary education preparation would be expected to occur (per IDEA).

Except for those with 504 plans, all other groups of youth included in this study had a higher percentage of male students compared to female students. We combined racial groups with low rates of occurrence into the larger group of White students for comparison to other groups. We found that across all groups, most youth were of White, Asian, or other racial groups compared to youth of Black (non-Hispanic) and Hispanic racial/ethnic groups. The majority of youth in this study were either sixteen or seventeen years old across all groups. Demographics of

the sample along with their confidence intervals are shown in Table 1.

Variables

To identify postsecondary education preparation activity variables, we reviewed the NLTS 2012 parent and youth surveys. We sought survey questions focusing on activities or services provided to youth during their transition experience that could be viewed as preparation for postsecondary education. Additionally, we reviewed existing research on postsecondary education preparation of youth with disabilities using NLTS-2 and NLTS 2012 (e.g., Qian et al., 2020; Trainor et al., 2019; Yu et al., 2018) to confirm our selection of variables aligned with previous research using this dataset. We identified eleven postsecondary education preparation variables (see Table 2). All variables were binary and were reported by either youth or parents. The design of the NLTS 2012 involved asking some questions to youth (e.g., receipt of academic supports, participation in IEP and transition planning, and employment experiences) and other questions to parents *about* their youth (e.g., disabilities and abilities, background characteristics, and type of school enrollment) with no overlap in the questions (Burghardt et al, 2017). Questions relevant to the present study were found in both youth and parent surveys, thus data from both sources were used in our analysis.

Data analysis

Data analysis was conducted using IBM SPSS software version 26, Complex Samples. We estimated the frequencies and the 95% confidence interval comparing the figures across the five groups of youth with intellectual disability and autism, autism, other disabilities, 504 plans, and no disabilities. To assess the magnitude of the differences across the five groups, we computed the effect size (*ES*) and applied Cohen's (1988) *h* standards to determine the strength of the effect. According to Cohen's *h* criteria, $ES < .20$ are considered negligible; $ES \geq .20$ are

small; $ES \geq .50$ are medium; and $ES \geq .80$ are large. Finally, we ran Chi-Square (χ^2) tests to identify the statistical significance of the differences across the five groups. All analyses were carried out using weighted data. All figures were rounded to the closest decimal in compliance with the Institute of Education Sciences (IES) confidentiality requirements for restricted-use data and clearance from IES was obtained to disseminate the findings.

Results

Research question 1 examined the extent to which youth with intellectual disability and autism are accessing postsecondary education preparation. Weighted estimates show nearly two-thirds of students with intellectual disability and autism experienced receiving guidance on classes to take to prepare for plans after high school (62%) and receiving help identifying possible career options (61%). The remaining nine postsecondary education preparation activities were experienced by fewer than one-third (and as low as five percent) of students with intellectual disability and autism. Table 2 shows the weighted estimates for postsecondary education preparation activities ordered by most to least frequently experienced by youth with intellectual disability and autism.

Research question 2 examined the extent to which youth with intellectual disability and autism are accessing postsecondary education preparation compared to youth in other groups. Significant differences were found between the extent to which youth with intellectual disability and autism accessed postsecondary education preparation activities compared to youth in other groups for all variables other than receiving help identifying possible career options.

Intellectual Disability and Autism vs. Autism

Comparing youth with intellectual disability and autism to autistic youth without intellectual disability, all comparisons had either a small or negligible effect size ranging from

0.02 to 0.23 (see note in Table 2 for effect size estimates).

Intellectual Disability and Autism vs. Other Disabilities

Comparing youth with intellectual disability and autism to youth with other disabilities, there was a medium effect size between the percentage who took PSAT, ACT, SAT, or community college placement tests (24% youth with intellectual disability and autism vs. 49% youth with other disabilities, Cohen's $h = .53$). The effect size for all other comparisons was either small or negligible (see Table 2).

Intellectual Disability and Autism vs. 504 Plan

Comparing youth with intellectual disability and autism to youth with 504 plans but no IEPs, there was a large effect size for the comparisons on two variables: PSAT, ACT, SAT, or community college placement tests taken (24% youth with intellectual disability and autism vs. 76% youth with 504 plan but no IEP, Cohen's $h = 1.09$) and postsecondary education credit for classes in high school (five percent of youth with intellectual disability and autism vs. 32% youth with 504 plan but no IEP, Cohen's $h = .75$). Two comparisons had a medium effect size: Youth worked for pay in past 12 months (32% youth with intellectual disability and autism vs. 66% youth with 504 plan but no IEP, Cohen's $h = .69$) and youth took classes to prepare for a career (25% youth with intellectual disability and autism vs. 52% youth with 504 plan but no IEP, Cohen's $h = .56$). All other comparisons were either small or negligible (see Table 2).

Intellectual Disability and Autism vs. No Disability

Finally, comparing youth with intellectual disability and autism to youth with no disability, there was a large effect size for the comparisons on two variables: PSAT, ACT, SAT, or community college placement tests taken (24% youth with intellectual disability and autism vs. 78% youth with no disability, Cohen's $h = 1.14$) and postsecondary education credit for classes in

high school (five percent of youth with intellectual disability and autism vs 38% youth with no disability, Cohen's $h = .88$). A medium effect size was found for comparisons on four variables: Youth worked for pay in past 12 months (32% youth with intellectual disability and autism vs 60% youth with no disability, Cohen's $h = .57$), youth took classes to prepare for a career (25% youth with intellectual disability and autism vs 56% youth with no disability, Cohen's $h = .64$), youth received help signing up for postsecondary education entrance tests (ten percent of youth with intellectual disability and autism vs 34% youth with no disability, Cohen's $h = .60$), and youth received help with financial aid (eight percent of youth with intellectual disability and autism vs 31% youth with no disability, Cohen's $h = .61$). The effect sizes for the remaining five comparisons were either small or negligible (see Table 2).

Discussion

These findings indicate relatively low access to most postsecondary education preparation activities for youth with intellectual disability and autism. Nine of the eleven postsecondary education preparation variables identified in the NLTS 2012 dataset were experienced by fewer than one-third of youth with intellectual disability and autism. Stark differences were seen when comparing youth with intellectual disability and autism to youth with 504 plans and youth with no 504 plans or IEPs on the extent to which they experienced postsecondary education preparation activities. However, it is worth noting some of the postsecondary education preparation activities (e.g., receiving postsecondary education credit for courses in high school) were not widely experienced even by youth with neither 504 plans nor IEPs. In the following sections, we 1) discuss the key findings of the analysis, 2) describe some of the limitations of the study, and 3) offer recommendations for practice and research.

Key Findings

Postsecondary Education Exploration and Planning Activities

Creating a postsecondary education-going culture in secondary education is critical to setting the expectation that postsecondary education can be an option for any interested and motivated high school student. This can be done by implementing practices promoting schoolwide attention to postsecondary education-going for non-dominant youth (Athanasides et al., 2016). One part of creating this postsecondary education-going culture is to include students with disability, including those with intellectual disability and autism, in regularly scheduled postsecondary education exploration activities including attending postsecondary education fairs, visiting postsecondary education institutions, taking placement and standardized tests, and completing financial aid and postsecondary education applications (Lalor et al., 2018). Our finding reflects only half of the youth without IEPs and 504 plans, as well as those with 504 plans, received help from school staff to attend postsecondary education fairs or participate in postsecondary education visits. Just over a quarter of the youth with intellectual disability and autism and autistic youth without intellectual disability received school staff support to access these postsecondary education exploration activities. The College Board's National Office for School Counselor Advocacy (NOSCA, 2012) states "school communities that intentionally encourage high aspirations for all students are more likely to help them gain the academic preparation necessary to graduate college and career ready" (p. 4).

Students with disabilities and their families, especially those with more significant academic support needs such as intellectual disability and autism, are often encouraged during transition planning talks to attend *transition resource fairs* to explore their future options. However, they are less likely to be invited or encouraged to attend to the *postsecondary education information events* by their educators, transition specialists, and high school

counselors (Grigal and Papay, 2020). Excluding parents and students from these events sends an explicit message that postsecondary education is not an option for these youth. While transition fairs offer important information to students with disabilities and their families about adult service and support from state and community agencies; rarely are postsecondary education institution representatives present for students to learn more about those post-school outcomes at these events (Grigal and Papay, 2020).

Postsecondary education visits are potentially a more impactful postsecondary education preparation activity as they can help the student see the campus and speak with current students and faculty. Campus visits, especially those conducted early in students' academic careers, may also impact future postsecondary enrollment rates (Smith et al., 2022). One study of eighth-grade students who visited a postsecondary education institution found they were more likely to exhibit higher levels of postsecondary education knowledge, efficacy, and grit, and had a higher likelihood of conversing with school personnel about postsecondary education than those who simply received information packets about the postsecondary education institution (Swanson et al., 2021). Although both exploration activities, postsecondary education fairs and postsecondary education visits, have a role in transition planning, students with intellectual disability and autism and autistic youth without intellectual disability may not be receiving the same opportunities as their peers to explore postsecondary education as a potential post-school outcome.

Standardized tests such as the PSAT, SAT, or ACT are used for postsecondary education admission as well as course placement for many prospective postsecondary education students. Recently, more postsecondary education institutions have made these tests optional by not requiring applicants to provide their test scores (Schultz and Backstrom, 2021). For students

seeking to take these exams, knowledge of the administration schedule, available accommodations, and the process through which to request them is important (Lalor et al., 2018). Our findings reflect the youth who most frequently took these tests were those with no 504 plans or IEPs (78%) and those with 504 plans (76%). Almost half (49%) of the youth with other disabilities were reported to have taken a placement test. The percentages for youth with intellectual disability and autism and autistic youth without intellectual disability were substantially lower (24% and 31%, respectively).

A very large effect size punctuated the difference between access for these youth and their counterparts with different disability statuses or youth with no disability. These results align with earlier studies that found youth with learning disabilities, speech, or other high-incidence disabilities were more likely to engage in exam preparation than those with autism, intellectual disability, or multiple disabilities (Wagner et al., 2003).

The development of new postsecondary education programs designed to support students with intellectual disability and autism creates an important caveat when considering standardized tests and placement tests. Students seeking admission to these programs typically apply via a separate application process than do other students, and standardized test scores are not a part of the application (Grigal et al., 2022). Therefore, we might not expect students in our target groups to take those tests as part of their postsecondary education preparation. However, given the time when these data were collected (2012-2013), it is doubtful the youth with intellectual disability and autism and autistic youth without intellectual disability who did not take placement tests refrained from doing so knowing they were not relevant to the application process.

The postsecondary education application process for all students, regardless of their disability status, can be overwhelming. The need for students, families, and educators to discuss

options and plan early is frequently noted in postsecondary education preparation materials regardless of disability status. For students with disability, Lalor et al. (2018) suggest planning begin as early as ninth grade. Our findings reflect about 50% of youth with no IEPs or 504 plans receive support on postsecondary education applications and slightly less than one-third receive support from staff with financial aid forms. For youth with other disabilities, the percentages are slightly lower for postsecondary education applications and substantially lower for financial aid support. These findings may reflect a more systemic lack of college preparation for all students, and this deficit becomes more exacerbated for students with more significant support needs.

The percent of youth with intellectual disability and autism and autistic youth without intellectual disability receiving support to complete postsecondary education applications were substantially lower than their counterparts with other or no disabilities. Most disturbing was the finding only eight and nine percent of youth with intellectual disability and autism and autistic youth without intellectual disability received support from staff to complete financial aid forms. This is especially troubling as access to financial aid for youth with intellectual disability and autism has expanded significantly since the HEOA was passed in 2008 (Grigal et al., 2021a).

High School Courses

Though youth with 504 plans and youth with neither 504 plans nor IEPs took courses at their high school for postsecondary education credit at far greater rates (e.g., 33 percentage points higher) than youth with intellectual disability and autism, there was overall low engagement in this postsecondary education preparation activity. Song et al. (2021) found students who take courses for which they earned postsecondary education credit during high school were more likely to enroll in two-year colleges and complete a postsecondary associate or bachelor's degree or certificate program of interest. Considering the known, positive, and long-

term impact on postsecondary outcomes for youth who receive postsecondary education credit for courses taken in high school (An et al., 2019), it is important for all youth to have access and opportunity to engage in early postsecondary education opportunities.

Data regarding the type of courses youth took in high school to prepare for a career of interest is limiting as we cannot discern the kind of courses youth were enrolled in. For example, some youth may have taken career and technical courses designed for specific vocations, whereas others may have taken segregated vocational skills courses more suited for youth seeking to gain pre-requisite skills needed for most vocations. Results found taking courses to prepare for a career was reported for between 25% and 56% of youth, suggesting this is an emerging area of postsecondary education preparation for all youth. However, without further knowledge about the names of courses and their related industry, it is difficult to ascertain the impact these courses had on preparing youth with and without intellectual disability and autism for postsecondary success.

Employment as a Postsecondary Education Preparation Activity

Youth with intellectual disability and autism in the present sample were only about half as likely as their peers with 504 plans or neither 504 plans nor IEPs to have worked for pay in the past 12 months. This is a disappointing finding given that paid employment while enrolled in high school has been found to be one of the most important research-based predictors of both post-school education as well as post-school employment (Mazzotti et al., 2021). As far back as 1991, the first NLTS study found only 40% of youth with intellectual disability to have had any type of paid job in the preceding year (Wagner et al., 1992) and we would hope to have seen progress in this critical transition experience. Early paid work experiences allow youth to see the value of work and learn the importance of education and training for achieving their career goals.

Access for youth with intellectual disability and autism to this important predictor of post-school outcomes is an area in need of much improvement.

Limitations

Several limitations of the present study must be acknowledged. Although the NLTS 2012 dataset is the most recent national data available to answer the research questions, these data were collected a decade ago, and therefore may not represent the present state of access to postsecondary education preparation activities by youth with intellectual disability and autism or youth in other groups. It is, however, the most recent national dataset on transition experiences of youth with disability and it offers the opportunity to examine the provision of services differentiated by disability type and status. Despite its age, findings from the NLTS 2012 continue to provide valuable insight to the transition field with at least 20 published studies utilizing the dataset since 2019. Among other findings, these studies have provided increased understanding of parent involvement in the transition process (see: Qian et al., 2020; Zeng et al., 2022), transition experiences and outcomes for English learners with disabilities (see: Trainor et al., 2022; Wu et al., 2022), and student participation in transition planning (see: Davenport et al., 2022; Bouck et al., 2021). Notably, seven of the publications focused on youth with intellectual disability and five on autistic youth. Given that these studies span across 15 different journals and were conducted by leaders in the transition field, it is prudent that information continues to be extracted from the dataset as it is our best (and only) representation of transition preparation practices experienced by most youth with disabilities at a national scale.

We were limited by the variables available in the dataset and could not ascertain the quality of services through the data analysis. Data were reported by youth and parents without triangulation by teachers or other transition professionals. It is possible that youth and their

parents may not have been aware of all postsecondary education preparation offered or delivered by these education professionals. Youth also may have chosen not to participate in the preparation activities described. We conducted a descriptive analysis to look for differences between groups based on disability type, but it is possible there are further explanations for group differences beyond disability type. Additionally, we defined intellectual disability and autism as youth who had either a primary diagnosis of intellectual disability or a primary diagnosis of autism coupled with a diagnosis of intellectual disability. There has been variability in how youth with intellectual disability are grouped with autistic youth in prior research, and therefore our findings may not be comparable to some existing studies. Finally, the dataset supplied no data on post-school outcomes that would have enabled us to better examine the relationship between postsecondary education preparation activities and actual post-secondary education enrollment.

Recommendations for Practice and Research

Practice

To help students with intellectual disability and autism engage in postsecondary education preparation activities, transition professionals must include it as a realistic choice on the list of potential post-school outcomes when conducting transition planning. Secondary educators, transition professionals, and high school counselors may be unaware of the current postsecondary education options available to students with intellectual disability and autism; especially if there are few in their geographic region. If education professionals have not learned about these options in their preservice or in-service training, the only mechanism for obtaining first-hand knowledge may come from working directly with students and their families who are planning for college. The concept of seeing is believing is relevant here; and transition and other

education professionals may need to contact or visit postsecondary education programs enrolling students with intellectual disability and autism to learn more about the students enrolled and resulting credentials and outcomes. Armed with this knowledge, educators can improve postsecondary education planning by including information about these options in their transition planning activities and postsecondary education fairs (Grigal et al., 2020).

Students, and in large part, their families, base their desired future goals largely on what they are told is possible by their teachers (Madaus et al., 2018). If the educators in charge of their planning indicate, either explicitly or implicitly, that postsecondary education is not an option, postsecondary education preparation will be less likely to occur. Families and students can be informed about online postsecondary education search resources available from national centers such as the College Autism Network and Think College and suggest visits to postsecondary education institutions to learn more about these programs. Parent information centers and parent advocacy

To effectively guide transition-age youth with intellectual disability and autism and autistic youth without intellectual disability in pursuing postsecondary education as an option, transition professionals must learn and share information about the application procedures and access to federal student aid. There are currently 168 postsecondary education institutions with approved Comprehensive Transition and Postsecondary (CTP) programs in the U.S. that can offer eligible students with intellectual disability access to certain forms of federal student grants and work-study funds (Federal Student Aid, 2023). Students in CTP programs do not need to have received a high school diploma or have a plan to matriculate toward a degree to qualify for federal student aid (Weir & Boyle, 2020). Without knowledge of current options, education, and counseling professionals are unlikely to guide students and families to use these resources.

An emerging model of transition service delivery that might support enhanced postsecondary education preparation for students with intellectual disability and autism is college-based transition services. This approach provides students between the ages of 18 and 22 with their transition services on a postsecondary education campus, allowing them to enroll in courses and participate in clubs and other organizations (Paiewonsky and Shahrzad, 2020). College-based transition programs also have a strong focus on supporting students to engage in employment and work-based learning (Papay et al., 2023). There are 108 postsecondary education institutions in the U.S. offering college-based services to transitioning youth (Think College, 2022). In some states like Massachusetts, the state's Department of Higher Education offers grants to postsecondary education institutions to develop these partnerships (Paiewonsky and Shahrzad, 2020). This approach requires significant collaboration between local education agency staff, postsecondary education staff, and other community agencies (Hanson, 2019). It facilitates early and consistent collaboration between these critical education partners and allows students to experience postsecondary education while receiving the extensive support allowed under IDEA-funded services. Secondary transition educators could explore and learn more about these options to support development and implementation in their local communities.

Research

The present study provides a picture of the postsecondary education preparation experiences of youth with intellectual disability and autism and youth with other disabilities. The NLTS 2012 offers national data on several transition-related variables, and future research could use this dataset to further disentangle the postsecondary education preparation experiences of youth considering additional variables beyond disability. Future iterations of the NLTS need to be designed with an eye to capturing a fuller picture of postsecondary education preparation

experiences of youth with disabilities. Data on high school course taking of these youth, along with data gathered from transition professionals on the types of postsecondary education preparation activities provided, would corroborate survey data from youth and parents and offer a richer source for research. Qualitative research with youth with intellectual disability and autism who ultimately did and did not go on to enroll in postsecondary education may lead to a better understanding of what it takes to prepare students with intellectual disability and autism for postsecondary education.

Without data on postsecondary education enrollment and completion of youth with disabilities, it is challenging to evaluate the impact of postsecondary education preparation activities provided during high school. Although there are several national databases that capture data on students in postsecondary education (e.g., Beginning Postsecondary Students Longitudinal Study, Integrated Postsecondary Education Data System, National Postsecondary Student Aid Study, National Survey of Student Engagement, and Postsecondary Education Quick Information System), many of which contain at least some disability-related variables inconsistencies in definitions of disabilities (Avellone and Scott, 2017) and the lack of recognition of postsecondary education-going students with intellectual disability and autism in these databases results in little usable data for understanding the postsecondary education enrollment experiences of youth with intellectual disability and autism. Giving greater attention to disability-related variables in the design of large-scale datasets on postsecondary education would further our ability to understand the importance and relative effectiveness of postsecondary education preparation activities for youth across disability groups.

Finally, expanded research is needed to determine if college-based transition services result in improved postsecondary education preparation for transitioning youth with intellectual

disability and autism. One recent study documented students taking part in a model of college-based transition programs had higher self-determination scores at the end of the school year than did students engaged in traditional transition services (non-college-based; Schillaci et al., 2021). Although recent federal guidance has been issued about the use of funds to support these experiences in secondary and postsecondary education (U.S. Department of Education, 2019), there is a great disparity in how and where these programs are provided. Better and more robust data are needed to support future research on the prevalence and impact of college-based transition services for students with intellectual disability and autism.

Conclusion

The findings from this study highlight points of stagnation in the postsecondary education preparation activities of students with intellectual disability and autism. Past low engagement rates in postsecondary education preparation activities may have been attributed to the limited access youth with intellectual disability and autism have had to postsecondary education programs. However, even with the rise of inclusive postsecondary education programs on postsecondary education campuses across the country and the creation of TPSID model demonstration programs specifically designed for youth with intellectual disability, youth engagement in postsecondary education preparation activities is still significantly low for those with intellectual disability and autism. With increased access and opportunities to pursue further education after high school, youth with intellectual disability and autism need to engage in more postsecondary education preparation activities at levels far surpassing previous years.

Moreover, while the information gained from national data and past research indicate current levels of engagement, little, if any, information is available about the quality of services youth with intellectual disability and autism receive. Detailed information is needed about the

level of inclusion with other youth (with and without IEPs and 504 plans) when engaged in activities and the connection between postsecondary education preparation and eventual postsecondary education enrollment and completion. Transition planning for students with intellectual disability and autism must address postsecondary education preparation in concrete, achievable ways to ensure these youth and their families can envision a future where postsecondary education is a realistic and attainable postsecondary option.

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Table 1

Demographic Characteristics

		Intellectual Disability	Autism	Other Disabilities	504 Plan	No Disability
		<i>N</i> = 1240	<i>N</i> = 680	<i>N</i> = 6,230	<i>N</i> = 490	<i>N</i> = 1,360
		% (95% CI)	% (95% CI)	% (95% CI)	% (95% CI)	% (95% CI)
Gender						
	Female	42 (38, 47)	18 (14, 22)	31 (29, 34)	39 (32, 46)	56 (51, 60)
	Male	58 (53, 62)	82 (78, 86)	69 (66, 71)	61 (54, 68)	44 (40, 49)
Race/Ethnicity						
	Black (Not Hispanic)	27 (22, 33)	11 (7, 15)	19 (16, 22)	16 (10, 24)	13 (10, 17)
	Hispanic	19 (15, 24)	14 (11, 19)	24 (20, 28)	12 (8, 17)	26 (21, 30)
	White, Asian, or other race	54 (48, 59)	75 (70, 80)	57 (53, 62)	72 (63, 80)	61 (56, 66)
Age						
	16	26 (22, 30)	29 (24, 34)	38 (35, 41)	51 (43, 59)	48 (43, 53)
	17	24 (21, 28)	29 (24, 34)	35 (32, 38)	33 (26, 41)	37 (33, 42)

18	25 (21, 29)	20 (16, 25)	20 (18, 22)	15 (10, 22)	14 (11, 17)
19	13 (10, 15)	12 (8, 17)	5 (4, 6)	0 (0, 0)	1 (0, 3)
20	8 (6, 10)	6 (4, 9)	2 (1, 3)	0 (0, 0)	0 (0, 0)
21	5 (4, 7)	4 (2, 8)	0 (0, 0)	0 (0, 0)	0 (0, 0)

Note. Confidence Intervals (CI) are provided for weighted sample estimates.

All figures were rounded to the closest decimal in compliance with the Institute of Education Sciences (IES) confidentiality requirements for restricted-use data, as such total percentage of youth characteristics may be +/- 100%.

Table 2

Estimates for College Preparation Variables

	Intellectual Disability	Autism		Other disabilities		504 plan		No disability		
	% (95% CI)	% (95% CI)	<i>ES</i>	% (95% CI)	<i>ES</i>	% (95% CI)	<i>ES</i>	% (95% CI)	<i>ES</i>	χ^2
Received guidance on classes to take to prepare for plans after high school	62 (57, 66)	63 (57, 69)	0.02	66 (63, 69)	0.08	70 (61, 77)	0.17	72 (67, 76)	0.21	9.1*
Received help identifying possible career options	61 (54, 67)	71 (63, 77)	0.21	73 (69, 76)	0.26	65 (54, 74)	0.08	70 (65, 75)	0.19	3.3
Youth worked for pay in past 12 months	32 (27, 37)	25 (20, 30)	0.16	47 (43, 51)	0.29	66 (57, 74)	0.69	60 (55, 66)	0.57	66.6**

Help reviewing results of college placement tests	32 (27, 39)	22 (17, 28)	0.23	46 (42, 49)	0.29	49 (40, 58)	0.35	54 (49, 60)	0.45	20.8**
Help completing college applications	29 (23, 35)	26 (21, 33)	0.07	40 (36, 44)	0.23	50 (40, 60)	0.43	51 (45, 56)	0.45	21**
College visits/fairs	28 (22, 36)	24 (18, 30)	0.09	41 (38, 45)	0.28	36 (27, 45)	0.17	51 (45, 56)	0.48	24.1**
Youth took classes to prepare for a career	25 (21, 30)	32 (27, 38)	0.16	41 (38, 45)	0.34	52 (44, 61)	0.56	56 (51, 62)	0.64	60.7**
PSAT, ACT, SAT, or community college	24 (20, 28)	31 (26, 37)	0.16	49 (45, 52)	0.53	76 (68, 83)	1.09	78 (74, 82)	1.14	350.9**

placement tests

taken

Help signing up	10 (7, 15)	13 (9, 17)	0.09	21 (18, 24)	0.31	25 (18, 33)	0.4	34 (29, 39)	0.60	48.4**
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for college

entrance tests

Help with	8 (6, 12)	9 (6, 14)	0.04	15 (12, 18)	0.22	17 (11, 25)	0.28	31 (26, 37)	0.61	54**
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financial aid

Postsecondary	5 (4, 8)	9 (6, 4)	0.16	13 (11, 15)	0.29	32 (24, 41)	0.75	38 (33, 44)	0.88	135.9**
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education credit

for classes in

high school

Notes. Cohen's *h* effect size (*ES*) is noted for each group in comparison with the intellectual disability group: Small *ES* =>.20;

Medium *ES*=>.50; Large *ES*=>.80.

* indicates a *p*-value of < .05; ** indicates a *p*-value of < .001.